

LeafyLand v2 — Foundation Phase

Identity, Vendor, Product Marketplace, and Orders bounded contexts — architecture, build log, and verification summary.

Repository leafylandmain Branch main Period Jul 11 – Jul 12, 2026

Summary

Following the foundation-phase documentation suite (Volumes 00–07: constitution, business requirements, SRS, system architecture, database design, and API specifications), five vertical slices were built, independently verified against a live PostgreSQL database, and committed. Each slice implements one bounded context end to end — real schema, real business rules, a working API, and a browser-reachable frontend page — rather than a partial or mocked implementation.

5 BOUNDED CONTEXTS	103 PASSING TESTS	14 FRONTEND ROUTES	8 COMMITTS TO MAIN
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Modules built

Identity

COMPLETE SLICE

Registration, email verification, login, JWT refresh with session-family theft detection, and protected profile reads.

Endpoints: 5 Key rule: refresh-token reuse revokes the entire session family (BR-ID-02)

Pages:

/register

/verify-email

/login

/account

Vendor

COMPLETE SLICE

Vendor registration, admin verification (approve / reject), store-profile self-management, and a minimal certification-document workflow.

Endpoints: **7** Authorization: DB-backed role lookup, scoped per vendor

Pages: `/vendor-register` `/dashboard`

Product Marketplace

COMPLETE SLICE

Categories, vendor-owned products and variants, publish/delist lifecycle, platform-wide SKU uniqueness, and public catalog browsing. Includes the organic-claim rule: a product can only be marked organic with an admin-approved certification on file, cross-checked against the Vendor module.

Endpoints: **13** Cross-context reads: 2 (Vendor status, certification documents)

Pages: `/products` `/products/[id]` `/vendor-products`

Orders

COMPLETE SLICE

Persistent cart, checkout with live price/stock re-validation, and a payment gateway with real HMAC-SHA256 webhook signature verification — the only code path permitted to mark an order paid.

Endpoints: **9** Key rule: order status changes only via a verified webhook (BR-ORD-01)

Pages: `/cart` `/checkout`

Reviews & Q&A

COMPLETE SLICE

Verified-purchase-only product reviews and public question-and-answer threads, with answers restricted to the product's owning vendor.

Endpoints: **5** Key rule: a review requires a paid order line the reviewer owns

Pages: reviews and Q&A embedded in `/products/[id]`

Notable finding: webhook signature bypass

CAUGHT DURING INDEPENDENT VERIFICATION, FIXED SAME SESSION

The Orders module's payment webhook is the only path allowed to mark an order paid, gated on a cryptographic signature check. Independent testing found that an incorrectly-shaped signature — a valid signature with one character appended — was still accepted. The cause: Node's `Buffer.from(str, 'hex')` silently truncates malformed hex instead of rejecting it, so the tampered string decoded to the same bytes as the correct one.

Fixed by validating the signature is exactly 64 hex characters before any decoding takes place. Re-verified live against the running server: the exact bypass that previously succeeded now correctly returns `401`, and the legitimate payment path was re-confirmed unaffected. Two regression tests were added to guard against recurrence.

Build log

DATE	COMMIT	CHANGE
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Jul 11	4475c8c	Foundation-phase documentation suite (Volumes 00–05)
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Jul 11	186a6e5	Volume 07 API Specifications
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Jul 11	3ba02e7	Identity + User vertical slice — monorepo scaffold
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Jul 11	2a7ea6d	Register and verify-email pages complete the auth flow
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Jul 11	36dc2c6	Vendor module; auth-store hydration race fixed
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Jul 11	494e593	Product Marketplace module; minimal vendor documents
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Jul 11	7ca84f1	Orders module — cart, checkout, webhook-verified payments
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Jul 12	5657325	Reviews & Q&A for Product Marketplace
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Verification approach

Every module was proven against a live PostgreSQL database and a running server, not just unit tests: registering real accounts, driving each business rule through both its accept and reject paths, and confirming the resulting database state directly. Where a slice was built by a delegated process, results were independently re-run rather than accepted on report — this is how the webhook signature gap above was found.

Explicitly deferred

Scoped out of the foundation phase to keep each slice a provable, working vertical rather than a broad partial surface. Tracked against the SRS for a later pass:

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| —OTP, OAuth, and MFA login | —Vendor staff invitations and wallet/commissions |
| —Product search, faceted filtering, and recommendations | —Offers, coupons, and brand catalog management |
| —Invoices, shipping/fulfillment, and returns | —Real Razorpay integration (order creation is stubbed) |
| —Guest cart and cart merge-on-login | —Full state-based GST calculation |